

CS 791: Practice questions on diffusion in discrete steps

1. In a diffusion model, assume that the initial distribution $q(x_0)$ is a mixture of two Gaussian distributions defined as

$$q(x_0) = 0.5\mathcal{N}(x_0; 2, 1) + 0.5\mathcal{N}(x_0; -1, 3)$$

Assume $\beta_t = 0.01, \forall t$ and number of steps $T = 100$.

- (a) Write the expression for $q(x_t|x_0)$ in the simplest possible terms for $t = 2$. ..2 Simple application of the formula discussed in class about direct jump
- (b) Write the expression for $q(x_t)$ in the simplest possible term where you work out the results of any integrals. ..4

Since the starting distribution is a mixture of two Gaussian. On integrating $\int_{x_0} q(x_t|x_0)q(x_0)$ we get a mixture. To simplify this integration we use the following result.

If x, y are two r.v. such that $P(x) \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $P(y|x) = \mathcal{N}(y; ax, \sigma_y^2)$, then $y = ax + \epsilon_2 = a\mu_x + a\epsilon_1 + \epsilon_2$ where $\epsilon_1 \sim \mathcal{N}(0, \sigma_x^2), \epsilon_2 \sim \mathcal{N}(0, \sigma_y^2)$. Since the sum of two Gaussian distributed R.V.s is Gaussian, we get that $y \sim \mathcal{N}(; a\mu_x; a^2\sigma_x^2 + \sigma_y^2)$.

This formula has to be plugged into the answer derived in the first part.

- (c) If in the forward phase, we sample t from a uniform distribution between a and b , that is $t = \mathcal{U}(a, b)$ what finite values of a, b do you recommend? ..1
 $a = 0, b$ can be any large values say 20 for which the $e^{-t^2/4}$ is close to zero.

2. Consider the diffusion model for generating a discrete sequence $\mathbf{x}$ of L words $x_1, \dots, x_L$ where each word can take one of m possible values. Assume, in the forward process each word j in the starting sequence can be independently masked as per this noising process:

$$q(x_j(t)|x_j(0)) = \begin{cases} [\text{mask}] & \text{with probability } 1 - \alpha_t \\ x_j(0) & \text{with probability } \alpha_t \end{cases}$$

- (a) What is the forward probability of all words in $\mathbf{x}(t)$ to be masked at time t ? ..1
- (b) Assume for the reverse step we trained a model to predict the probability distribution $P_\theta(\mathbf{x}(s)|\mathbf{x}(t)) = \prod_j P_\theta(x_j(s)|\mathbf{x}(t))$ where each $P_\theta(x_j(s)|\mathbf{x}(t)) = q(x_j(s)|x_j(t), \mu_{\theta,j}(\mathbf{x}(t), t))$ as discussed in class, and $s = t - 1$ What is the probability that all L tokens in $\mathbf{x}(s)$ correspond to the first word, i.e, $\mathbf{x}(s) = \{1, \dots, 1\}$ when $\mathbf{x}(t)$ has all masked tokens. ..2 The probability that a masked word in step t is un-masked in step s is $\frac{\alpha_s - \alpha_t}{1 - \alpha_t}$. The probability that the specific token is unmask is token "1" is $\mu_{\theta,1}(\mathbf{x}(t), t)$. This means that the total probability is $(\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \mu_{\theta,1}(\mathbf{x}(t), t))^L$
- (c) In class, we did not explicitly write down the reverse sampling algorithm for discrete generation. Using the notation introduced above, write down that algorithm. Assume you are given α_t and you discretize time from 1 to 0 into T steps during generation. ..3

Start with $\mathbf{x}(1) = \text{all masks}$.

for $i = 0$ to T **do**

Let $t = 1 - i/T, s = t - 1/T$

Compute $\mu_{\theta,j}(\mathbf{x}(t), t)$ in one forward pass over the denoiser.

An existing masked word is chosen to be unmasked with probability $\frac{\alpha_s - \alpha_t}{1 - \alpha_t}$. If position j is selected for unmasking choose a word with probability $\mu_{\theta,j}(\mathbf{x}(t), t)$. Exit when all words are unmasked.

end for

- Standard diffusion models operate directly in high-dimensional spaces, making them computationally expensive and inefficient in terms of both speed and memory. One trick to mitigate these limitations is to perform diffusion in a low-dimensional latent space and then project the result back to the original space. This is known as Latent Diffusion.

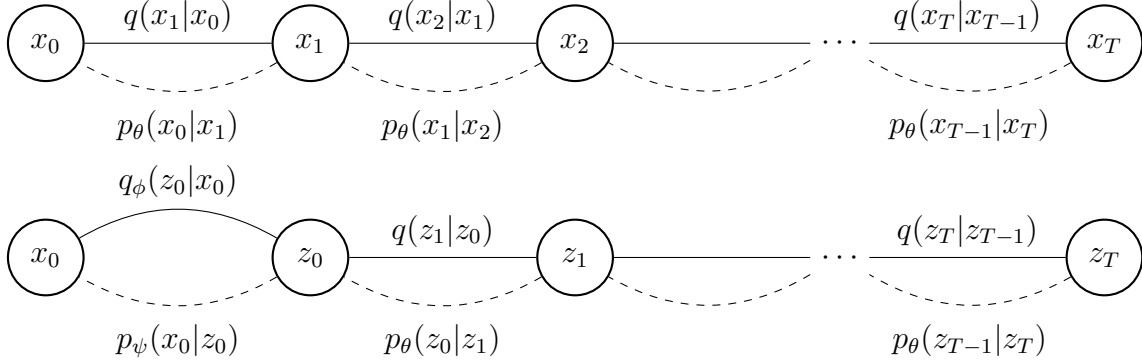


Figure 1: Standard Diffusion vs Latent Diffusion. x_0 is the data we want to generate

Note that now you have three sets of parameters: (ϕ) - An encoder network to project x_0 to latent vector z_0 , (ψ) - A decoder network to project z_0 to x_0 , and (θ) - the denoiser network.

- Derive the training objective using the variational approximation, expressing it in terms of q_ϕ , p_ψ , and p_θ . For your reference the corresponding objective for standard diffusion is:

$$\mathbb{E}_q \left[D_{\text{KL}}(q(x_T|x_0) \| p(x_T)) + \sum_{t>1} D_{\text{KL}}(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t)) - \log p_\theta(x_0|x_1) \right]$$

You need to modify the above for latent diffusion. ..3 Trick is to first derive the VAE objective,

$$\mathcal{L}(\theta, \phi, \psi) = \mathbb{E}_{x_0} [\mathbb{E}_{z \sim q_\phi(z|x)} [-\log p_\psi(x|z)] + D_{\text{KL}}(q_\phi(z|x) \| p_\theta(z))]$$

and interpret $z = [z_0, z_1, \dots, z_T]$. This interpretation allowed us to rewrite $p(z)$ as $p_\theta(z)$ and then we can proceed with factorising $P_\theta(z)$ as $P(z_T) \prod_{t \in [1, \dots, T]} P_\theta(z_{t-1}|z_t)$

- Simplify the objective to obtain a loss function on a dataset D . You are allowed to assume appropriate forms of q_ϕ , p_ψ , and p_θ . Again, corresponding objective for Standard diffusion is,

$$\mathcal{L}(\theta) := \sum_{x_0 \in D} \mathbb{E}_{t, \epsilon} \left[\left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$

You are encouraged to use similar notations. ..3 After assuming proper parameterisation on of all the distributions, we will get weighted denosing objective + reconstrucion terms. Which need to simplified further to obtain a simpler looking loss functions

- (c) Describe a practical training algorithm to jointly optimize ϕ, ψ, θ . For reference the standard diffusion training algorithm is:

Sample $x_0 \sim q(x_0)$ Sample $t \sim \text{Uniform}(\{1, \dots, T\})$ Sample $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ Take gradient descent step on

$$\nabla_{\theta} \left\| \epsilon - \epsilon_{\theta} \left(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t \right) \right\|^2$$

converged

..3

- (d) Now, consider a two-stage training process, where the encoder (ϕ) - decoder (ψ) networks are trained independently of the denoiser network to learn a latent space. Once trained, the encoder is used to map the entire dataset into latent vectors, on which the diffusion model is then trained. Discuss the advantages and disadvantages of this approach compared to joint training described earlier.

..1 **End-to-end latent diffusion training** offers better sample quality through joint optimization but requires more computational resources and suffers from training instability, while the two-stage approach (separate autoencoder and diffusion training) provides modularity, easier debugging, and efficiency at the cost of potential information loss

- (e) Compare the use of a VAE versus using a simple autoencoder for generating the dataset of latent vectors. Keep in mind that a simple autoencoder optimizes only for reconstruction loss (objective provided). Discuss the potential impact on diffusion performance trained on this dataset.

$$\mathcal{L}_{\text{AE}}(\theta, \phi) = \mathbb{E}_{x \sim q(x)} [\|x - D_{\theta}(E_{\phi}(x))\|^2]$$

..2 **KL term in VAEs** objective enforces that the latent embeddings follow a normal distribution while trading off reconstruction quality. AE on the other hand optimize only for reconstruction, and thus distribution of latent embeddings of training data can be arbitrarily complex. VAEs will make it easier to train diffusion model, but AE will lead to better quality samples for a well-trained diffusion model.